SplitSplit

Problem and Solution Proposal

Team members and pages

Komil Qodirov — Page 1

Mavluda Sotvoldiyeva — Pages 2–4

Nurlan Maxsudov — Page 5

Dilnavoz Sultonova — Page 6

SplitSplit

Problem statement and target users

1. The problem

Students who share rented apartments in Tashkent often pay for common items such as groceries, cleaning supplies and utilities. When different people pay at different times, it can be difficult to remember each payment and calculate how much everyone owes. SplitSplit focuses on making these shared expenses easier to track.

2. Who faces this problem?

Our target users are university students in Tashkent who live with roommates and regularly share household expenses. The proposed app is designed for small groups, typically two to five people.

3. How verified the problem

We survey students who currently share accommodation. The survey asks how they record expenses, how often they have trouble calculating balances, and what difficulties they have experienced.

Participants surveyed	12
Students reporting a difficulty	8
Most common current method	Message in telegram

4. Why SplitSplit ?

The app would let roommates record a shared purchase, choose who participated, and see automatically calculated balances. The goal is a clear record of shared costs, not online banking or money transfers.

SplitSplit

App overview and main user flow

1. What is SplitSplit ?

SplitSplit is a simple mobile app for students who share an apartment. Roommates can record purchases made for the household and see how much each person owes or should receive. The app tracks expenses in soums (UZS).

2. Main user flow

Step 1 — Open a group: The student opens their apartment group and sees recent shared expenses.

Step 2 — Add an expense: They enter the purchase name, amount, and who paid.

Step 3 — Choose roommates: They select which group members should share that expense equally.

Step 4 — View balances: SplitSplit calculates each person's share and updates the balances.

Step 5 — Record a repayment: After roommates exchange money outside the app, they record the repayment.

3. Simple example

Four roommates share a grocery expense of 200,000 UZS. One person pays the full amount. Each roommate's share is 50,000 UZS. SplitSplit records who paid and shows the amount others owe.

4. What we CAN do

Create a group, add members, record expenses, split expenses equally, show balances, and mark repayments. Payments themselves happen outside the app.

TEAM WINX / MOBILE PROGRAMMING MIDTERM

SplitSplit

Initial versions — screens 1 to 3

These are simple sketches of the planned app screens. They show the layout and actions, not final visual design.

1. My Groups

My Groups	‹ Group Dashboard	‹ Add Expense
My apartment	Total spent: 400,000 UZS	Name: [Groceries]
4 members	Your balance: +100,000	Amount: [200000]
[Open group]	Groceries 200,000	Paid by: [You v]
[+ New group]	Internet 80,000	Share with: ☑ A ☑ B ☑ C
View or create a roommate group.	[+ Add expense]	[Save expense]
	See expenses and balances.	*Enter a purchase and select participants.*

Navigation

My Groups → Group Dashboard → Add Expense. After saving an expense, the app returns to the Group Dashboard with updated totals.

Forms and validation

The Add Expense screen requires a purchase name, an amount greater than zero, a payer, and at least one participant. The Save button does not accept an incomplete form.

SplitSplit

Initial versions — screens 4 and 5

4. Settle Up

‹ Settle Up	‹ Members & Settings
You receive: 100,000 UZS	Group: My apartment
Roommate B owes you 50,000	• You
Roommate C owes you 50,000	• Roommate A
[Record repayment]	• Roommate B
See balances and record money paid back.	• Roommate C
	[+ Add member]
	See the group and manage its members.

How the screens connect

My Groups → Group Dashboard

Group Dashboard → Add Expense → Group Dashboard

Group Dashboard → Settle Up → Group Dashboard

Group Dashboard → Members & Settings → Group Dashboard

Navigation plan

Buttons open the next screen using Flutter navigation. The back button returns to the previous screen.

Design notes

Each screen has one main task. Expense details are kept simple, and all amounts are shown in UZS.

SplitSplit

Flutter implementation plan and data model

1. Flutter screens and widgets

Screen	Widgets and layout
My Groups	Scaffold, Column, ListView, Card, ElevatedButton
Group Dashboard	Scaffold, Column, ListView, Card, Text
Add Expense	Scaffold, Form, Column, TextFormField, CheckboxListTile, Button
Balances	Scaffold, Column, ListView, Text
Members	Scaffold, Column, ListView, TextFormField, Button

2. Navigation, state and validation

Navigation: Navigator.push() opens a screen, and Navigator.pop() goes back. From the dashboard, users can open Add Expense, Balances and Members.

State: Riverpod (NotifierProvider) holds groups, roommates and expenses in memory. When an expense is added, the app recalculates the displayed balances.

Validation: Group and roommate names cannot be empty. The expense form requires a name, a positive amount, a payer and at least one participant.

3. Simple Dart data model

Roommate	id, name
Group	id, name, members
Expense	id, description, amount, paidBy, sharedWith, groupId

4. Calculation example

If one roommate pays 200,000 UZS for groceries shared by four people, each share is 50,000 UZS. The other three each owe the payer 50,000 UZS. The app only displays amounts; it does not transfer money.

5. Limits of the first version

No user accounts, cloud synchronization, payment processing or permanent storage. All data is kept in memory while the app is running.

SplitSplit

Project scope and team contributions

1. What we plan to build

SplitSplit will help students living together track shared household expenses. Users can create a group, add roommates, enter an expense, select who shared it, and see how much each person owes or should receive. The app only calculates amounts. It does not transfer money.

2. What is outside our scope

The app will not include bank cards, online payments, user accounts, cloud syncing, notifications, or AI features. We will keep the data in memory while the app is running. Our focus is on simple Flutter screens and correct calculations.

3. Team contributions

Team member	Main responsibility
Komil Qodirov	App idea, Problem research, survey and evidence
Malvuda Sotvoldiyeva	App overview, user flow and screen sketches
Nurlan Maxsudov	Flutter implementation plan and Dart data model
Dilnavoz Sultonova	Document coordination, editing and presentation preparation

4. Research evidence appendix

Survey response link: Google sheets

5. Use of AI tools

We used ChatGPT to help organize and draft the document. The team reviewed and edited the final text and is responsible for the research, design decisions, and explanations.

Note: The proposed features and design represent our initial development plan. Some implementation details or interface elements may change during development based on testing and feedback, while the main purpose of SplitSplit will remain the same.